WEEK 8 · OCT 29 · RUN IT YOURSELVES

Answering Questions

Becca is away again. Today is one thing only: answering a committee's questions in one sentence, then stopping. On Oct 22 the pitches were clear and the answers ran long. Today we fix the answers.

Before class

Read the **Read first** page and the **Question bank**. Bring your phone, earbuds, and a highlighter.

In class: one group, the four of you

Min	Block	What happens
0–2	**Set up**	Phones out and recording app open. Agree who takes which role in the first round, and who keeps time.
2–30	**Hot seat**	Four rounds of 7 minutes, one per person. Order: Marcio, Carol, Young, Borna. **Record your own round** on your phone (Voice Memos or any recorder), set on the table in front of you.
30–60	**Listen and fix**	Everyone at once, with earbuds. This is the main event. Listen to your round, pick your three worst answers, and fix them by hand one at a time: write it out, highlight, rewrite, record again. **Two fixes at least, three if you can.**

Each round, 7 minutes

Min	Block	What happens
0–1	**Headline**	The presenter starts their recording, then says what their research found (or expects to find) in **one sentence**. No slides.
1–7	**Questions**	The other three each play one committee role (below; take a different role each round) and fire questions. Rapid, one question at a time. **Everyone asks at least one follow-up.** Askers tick their score card as each answer ends and hand it over when the round is done.

Three rules for the askers

Cut at 30 seconds

The person on the presenter's right is timekeeper. At 30 s, say "Time." Mid-sentence is fine. The presenter then gives the answer again **in one sentence**.

Make them answer

If the first sentence didn't answer the question, say: "So is that a yes or a no?" or "So what's the number?" Then wait.

Play your role

Each asker takes one role and asks from it; rotate roles every round. **Mechanics:** what is physically happening. **Model:** why this model, and is it right. **Data and users:** where the data came from, and who acts on the result. Example questions are on the **Question bank** page; start there, then follow up on what you hear.

After class: send Becca one PDF with your Listen and fix pages and the three score cards you got, plus your **last re-recording** (the best fixed answer).

WEEK 8 · READ FIRST

Answer, Why, Stop

A committee question is not a cue to give a mini-lecture. It is a question. Answer it.

1. Answer

The first sentence is the answer: yes, no, partly, a number, a name, or "I don't know yet."

2. Why

One sentence of reason or evidence. One, not three.

3. Stop

Stop talking. If they want more, they will ask. Silence after your answer is fine.

"DID YOU VALIDATE THE MODEL?" RAMBLING

"So, validation is a big topic. First I set up the mesh, and I did a convergence study, and then I looked at some papers that used similar models, and the material properties came from the literature, so I think in a way, yes, it's validated, more or less."

35 seconds and the listener still doesn't know.

ANSWER, WHY, STOP

"Partly. It matches a hand calculation within 10%, but I haven't compared it to a physical test."

6 seconds. If they care about the test, they will ask about it next.

A move for each kind of question

Examples use the class example from Weeks 5 and 6: composite strips on historic brick walls under tornado load. A: the bond to lime mortar controls. B: the strip's strength controls.

They ask	Your move	Sounds like
Yes or no "Does the strip ever break before it debonds?"	Yes or no first, then the why.	"No, not on lime mortar. The bond goes at about a third of the strip's strength."
What is it "What does your load represent physically?"	Plain words, no jargon. Say what it leaves out.	"The wind pressure from an EF2 to EF3 tornado on the long wall, held steady. It leaves out debris and the sudden pressure drop."
Why this, not that "Why not test cement mortar too?"	Name the choice and the reason. Don't apologize.	"Because on cement mortar the bond is strong, so A and B predict the same thing. Lime mortar is where they split."
Does it generalize "Would this work in an EF5?"	Say it's outside your claim, and what would change.	"I'm not claiming that. Above EF3 the wall fails before the strips matter, so it would need a different retrofit."
Challenge "Your model says the wall fails, but the real one stood."	Agree with what's true. Say what it means.	"Right, so the model is missing some strength. My best guess is the floor tying the wall in, and I'm adding it next."
You don't know	Say so, then how you'd find out.	"I don't know yet. I'd get it from the mortar tests in my materials data."
You didn't follow	Ask which one they mean. Then answer.	"Do you mean the strip or the bond?"

Take three seconds. Pausing to think looks confident; starting to talk before you know your answer is how rambling starts. **Answer the question, not the slide.** They almost never want your whole method again.

WEEK 8 · QUESTION BANK · FOR THE ASKERS

Ask Like a Committee

Pick your role for the round. Start from these, put in their words (their load, their model, their buildings), then follow up on whatever the answer leaves open.

Mechanics

What is physically happening?

- What does that load represent physically? What does it leave out?
- What fails first, and how do you know?
- Why is that boundary condition right for this building?
- Is that number big or small? Compared to what?
- If your model is right, what would we see on the real building?
- Where does the load go after it hits the wall?

Model

Why this model, and is it right?

- Why this model and not a hand calculation?
- What did you compare it to? How close was it?
- Which one assumption, if wrong, changes your answer?
- How much does the result move if you change the input you trust least?
- Your model says it fails, but the real one stood. Why?
- Can you finish this by December? What would you cut?

Data and users

Where did it come from, and who acts on it?

- Who uses this, and what decision does it change?
- What would an engineer or preservation officer do differently tomorrow?
- Where did that data come from? How good is it?
- How many buildings is that based on?
- Is your “truth” biased? Who collected it, and how?
- Does this hold for buildings that aren’t like yours? What data do you wish you had?

Follow-ups, any role

Use these when an answer runs long, drifts or dodges.

- So is that a yes or a no?
- What’s the number?
- Can you say that in one sentence?
- Why?
- How do you know?
- What result would change your mind?

Good questions are short, ask one thing, and can’t be answered by re-showing a slide. Ask why and what if, not only what.

WEEK 8 · LISTEN AND FIX · 30 MINUTES

Hear Yourself Answer

Name _____ You will hear things the askers were too polite to put on the card.

1. Listen to your whole round once and time every answer

Use the recording's timestamps. "Seconds until the answer" is how long you talked before you said the yes, no, number or name. Set-up and background count against you. Goal: 0 to 3 seconds. About 5 minutes.

#	The question (a few words)	Seconds until the answer	Total seconds	Worst 3
1				☐
2				☐
3				☐
4				☐
5				☐
6				☐
7				☐
8				☐
9				☐
10				☐

2. Tick your three worst

The longest waits before the answer, or answers that never answered. Fix them worst first, one per **Fix an Answer** page, about 8 minutes each:

1. **Write out** the whole answer by hand, word for word.
2. **Highlight** the fewest words that answer the question.
3. **Rewrite** starting from those words: answer, why, stop.
4. **Record** the new version and play it back.

Two fixes at least, three if you have time. Finish the second before starting the third.

WEEK 8 · LISTEN AND FIX · FIX 1 OF 3

Fix an Answer

Question # ____ from your timing page: _____

1. Write out the whole answer, word for word

Pause the recording as often as you need. Every “so,” “um,” repeat and restart goes in.

2. Highlight the fewest words that answer the question

In the box above. Often it is a few words in the middle or at the end. If nothing answers it, write “didn’t answer” and what the answer should have been.

3. Rewrite it from the highlighted words

Start with them. Then one sentence of why. Then stop.

4. Say it into your phone, play it back

Without reading it. Again until it is under 15 seconds and the first sentence answers.

Original: _____ s New: _____ s Tries: _____

What you noticed hearing yourself that you didn’t notice while talking:

Fix an Answer

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What you noticed hearing yourself that you didn’t notice while talking:

WEEK 8 · LISTEN AND FIX · FIX 3 OF 3

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WEEK 8 · SCORE CARD · ONE PER ASKER PER ROUND

How Did the Answers Go?

Asker _____ Presenter _____ My role: mechanics model data and users

One row per question you asked, follow-ups included. Tick each box that's true as the answer ends. Jot the question while they answer.

	The question (a few words)	1st sentence answered it	Under 30 s	Then stopped	Had to redo
1		☐	☐	☐	☐
2		☐	☐	☐	☐
3		☐	☐	☐	☐
4		☐	☐	☐	☐
5		☐	☐	☐	☐

Any answer that dodged the question? Which #, and what they answered instead. Tell them at the end of the round; it's a good one to fix.

WEEK 8 · SCORE CARD · ONE PER ASKER PER ROUND

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